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T.E. (Computer Engg.)

WEB TECHNOLOGY

(2019 Pattern) (Semester - II) (310252)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.

- Q1) a)** What is difference between server side scripting language and client side scripting language. **[5]**
- b)** Describe servlet architecture in detail. **[4]**
- c)** Explain DTD in XML with schemes, elements & attributes. **[9]**

OR

- Q2) a)** What is session? How cookies & URL rewriting for session management in servlet. **[9]**
- b)** Write short note on : **[9]**
- i) AJAX
 - ii) XML transformation

- Q3) a)** Explain life cycle of JSP. Write advantages of JSP over servlet. **[9]**
- b)** Write a short notes on : **[8]**
- i) SOAP
 - ii) WSDL

OR

- Q4) a)** Explain streets framework with its components. Also explain interceptors. **[9]**
- b)** Explain JSP support for MVC i.e. model, view controller for developing web application. **[8]**

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- Q5) a)** Identify and explain steps involved in connecting to MySQL with PHP. **[9]**
- b)** Write short notes on : **[9]**
- i) Overview of ASP. NET
 - ii) Overview of C#
- OR
- Q6) a)** Explain in detail WAP Architecture & WML. **[9]**
- b)** Explain functions in PHP with example & session management. **[9]**
- Q7) a)** Explain classes and objects in Ruby with appropriate examples. **[9]**
- b)** Introduce the concept of Rails application. Describe layouts & stylesheet in Rail. **[8]**
- OR
- Q8) a)** Explain the scalar types & their operations in Ruby. **[8]**
- b)** Explain Architecture of EJB & explain types of EJB in detail. **[9]**

Solution - The Vijay Academy

Q1 a) Difference between Server Side Scripting & Client Side Scripting (5 Marks)

Ans:

Server-Side Scripting

- Code runs on the **server**.
- It generates **dynamic content** based on database or logic.
- Output is sent to the client as **HTML**.
- The user cannot see the code.
- Examples: **PHP, Java Servlets, ASP.NET, Node.js**

Client-Side Scripting

- Code runs in the **user's browser**.
- Used for **validations**, animations, UI changes.
- The user **can see the script** using inspect element.
- Does not interact directly with database.
- Examples: **JavaScript, HTML, CSS**

Server-Side Scripting	Client-Side Scripting
Runs on server	Runs in browser
Generates dynamic content	Handles UI/validation
Cannot be viewed by user	Can be viewed by user
Slower (server process needed)	Faster (local execution)
Examples: Servlets, PHP	Examples: JavaScript

Q1 b) Describe Servlet Architecture in Detail (4 Marks)

Ans:

A **Servlet** is a Java program that runs on a **web server** and responds to client requests.

Servlet Architecture Components

1. Client (Browser)

- Sends HTTP request (GET/POST).

2. Web Server

- Receives request and forwards it to the servlet container.

3. Servlet Container (e.g., Tomcat)

- Manages servlet life cycle.
- Responsibilities:
 - Loading servlet
 - Creating servlet instance
 - Calling init(), service(), destroy()
 - Managing threads
 - Managing sessions and resources

4. Servlet

- Processes the request
- Generates dynamic output (HTML/JSON/XML)

5. Response

- Servlet sends response back → web server → browser.

Servlet Architecture Flow

Client → Web Server → Servlet Container → Servlet → Response

Servlet Life Cycle

1. **init()** – Called once when servlet loads
2. **service()** – Called for each request
3. **destroy()** – Called before servlet is removed

Key Features

- Platform independent
- Uses Java
- Supports sessions, cookies
- Efficient (multi-threaded)

Q1 c) Explain DTD in XML with Schema, Elements & Attributes (9 Marks)

Ans:

1. What is DTD?

DTD = **Document Type Definition**

It defines the **structure** and **rules** of an XML document.

It tells:

- Which **elements** are allowed
- Their **order**
- Their **attributes**
- What data they contain

DTD ensures the XML file is **valid**.

2. DTD Example

XML File:

```
<?xml version="1.0"?>
<!DOCTYPE student SYSTEM "student.dtd">
<Student>
    <Name>Vijay</Name>
    <Roll>23</Roll>
</Student>
```

DTD File (student.dtd):

```
<!ELEMENT Student (Name, Roll)>
<!ELEMENT Name (#PCDATA)>
<!ELEMENT Roll (#PCDATA)>
```

3. Elements in DTD

An **element** is an XML tag.

Syntax:

```
<!ELEMENT tagname content>
```

Examples:

- **(#PCDATA)** = Parsed Character Data (text)
- **EMPTY** = No content
- **ANY** = Can contain anything

Example:

```
<!ELEMENT Book (Title, Author)>
```

4. Attributes in DTD

Attributes provide **extra information** to elements.

Syntax:

```
<!ATTLIST element attribute type value>
```

Example:

```
<!ATTLIST Student id CDATA #REQUIRED>
```

Types:

- **CDATA** – normal text
- **ID** – special unique id
- **#REQUIRED** – must be present
- **#IMPLIED** – optional

5. Internal vs External DTD

Internal DTD

Defined inside XML file.

```
<!DOCTYPE Note [  
  <!ELEMENT Note (To, From)>  

```

External DTD

Defined in a separate file.

```
<!DOCTYPE Note SYSTEM "note.dtd">
```

6. XML Schema (XSD)

Schema is an improved version of DTD.

- Supports **data types** (int, string, date)
- More powerful and flexible

Example:

```
<xs:element name="age" type="xs:integer"/>
```

OR

Q2 a) What is Session? How cookies & URL rewriting are used for session management in Servlets? (9 Marks)

Ans:

1. What is a Session?

A **session** is a way to store **user-specific data** on the server for a limited time.

- When a user visits a website, a **session ID** is created.
- This session ID is used to **identify the same user** across multiple pages.
- Session helps in:
 - Login systems
 - Shopping carts
 - User preferences
 - Maintaining user activities

Session = Technique to remember the user until they leave or close the browser.

2. Session Management Using Cookies

What are Cookies?

- Small text files stored on the **client's browser**.
- Servlet uses cookies to store the **session ID**.

How cookies manage session:

1. When a user first visits, server generates a **session ID**.
2. Server sends this session ID to browser in a **cookie**.
3. Browser sends this cookie back with each request.
4. Server reads the session ID → identifies the user.

Example:

```
Cookie c = new Cookie("sessionID", "12345");  
res.addCookie(c);
```

Advantages:

- Automatic session tracking
- Easy to implement

Disadvantages:

- User may disable cookies
- Limited data can be stored

3. Session Management Using URL Rewriting

What is URL rewriting?

When cookies are disabled, session ID is added **inside the URL** itself.

Example:

```
http://example.com/home;jsessionid=ABC123
```

How it works:

1. Server generates session ID.
2. Session ID is appended with every URL the user clicks.
3. Server reads the ID from URL.
4. User session is maintained.

Code Example:

```
String url = res.encodeURL("home.jsp");
```

Q2 b) Write short notes on: AJAX & XML Transformation (9 Marks)

Ans:

i) AJAX (Asynchronous JavaScript and XML)

Meaning

AJAX is a technique used to update a webpage **without reloading** the entire page.

Full Form

Asynchronous
Java
AScript and
XML

How AJAX Works

1. Browser creates an XMLHttpRequest object.
2. Sends request to server **asynchronously**.
3. Server processes and sends **data only**, not full page.
4. JavaScript updates only the required part of the page.

Example

```
var xhttp = new XMLHttpRequest();
xhttp.onload = function() {
    document.getElementById("result").innerHTML = this.responseText;
}
xhttp.open("GET", "data.txt", true);
xhttp.send();
```

Advantages

- No page reload
- Faster and smoother UI
- Used in live search, chat apps, Gmail inbox refresh

Applications

- Form validation
- Loading comments dynamically

ii) XML Transformation

Meaning

XML Transformation means converting an XML document into:

- HTML
- Another XML
- Text
- PDF (via tools)

This is done using **XSLT (Extensible Stylesheet Language Transformations)**.

Why needed?

- XML is for data storage
- XSLT converts the raw XML into **readable output**

XSLT Components

1. **XSL** → stylesheet
2. **XPath** → selects XML nodes
3. **Templates** → describes how data should be displayed

Example XML

```
<student>
  <name>Vijay</name>
</student>
```

XSLT Example

```
<xsl:template match="student">
  <h1><xsl:value-of select="name"/></h1>
</xsl:template>
```

Process

1. XML + XSLT = HTML output
2. Browser or processor applies the stylesheet
3. Final transformed document is shown to user

Q3 (a) Explain life cycle of JSP. Write advantages of JSP over Servlets.

(9 Marks)

Ans:

A JSP (Java Server Page) is a web page that contains Java code + HTML.

Its life cycle is the series of steps that happen from the moment a JSP is requested until it is destroyed.

JSP Life Cycle

1. Translation Phase

- The JSP file (.jsp) is **converted into a Java Servlet file** by the JSP engine.
- Example: index.jsp → index_jsp.java.

2. Compilation Phase

- The generated .java servlet file is **compiled** into a .class file.
- Example: index_jsp.java → index_jsp.class.

3. Loading and Instantiation

- The compiled servlet .class file is **loaded into memory**.
- A **JSP object/instance** is created.

4. Initialization — jspInit()

- The JSP container calls jspInit() only **once**.
- It is used to write initialization code (like database connection).

5. Request Processing — _jspService()

- For **every request**, _jspService(request, response) is executed.
- This method:
 - Reads client input
 - Executes Java code inside the JSP
 - Generates the output HTML

6. Destruction — jspDestroy()

- Method is called when the JSP is removed from memory.
- Used to close resources (database, files, etc.)

Advantages of JSP over Servlet

1. Easier to Write

- JSP mixes HTML + Java easily.
- In Servlet, writing HTML inside Java code becomes messy.

2. Better Separation of Presentation & Logic

- JSP for UI
- Servlet for backend logic
- Makes MVC architecture easy.

3. Automatic Servlet Generation

- JSP is automatically converted into a servlet.
- No need to write full Java code manually.

4. Fast Development

- Changing JSP does **not require full recompilation**.
- Server auto-updates changes.

5. Built-in Objects Provided

- JSP provides implicit objects like:
 - request, response, session, application, out
- Makes coding easier.

6. Custom Tags & Tag Libraries

- JSTL makes coding more powerful and readable than Servlets.

Q3 (b) Write short notes on: i) SOAP ii) WSDL(8 Marks)

Ans:

i) SOAP (Simple Object Access Protocol)

Concept in Brief

SOAP is a **protocol** used to send messages between applications over the internet.

It uses **XML** for data exchange.

Explanation

1. SOAP is XML-based

- All data and messages are written in XML.
- Platform-independent and language-independent.

2. Uses Standard Internet Protocols

- Mostly uses **HTTP**
- Can also use SMTP or other protocols.

3. SOAP Structure

A SOAP message has:

- **Envelope** → root element
- **Header** → optional information
- **Body** → main message
- **Fault** → error details

4. Highly Secure & Reliable

- Supports WS-Security standards.
- Used in banking, telecom, enterprise applications.

5. Interoperable

- Any system (Java, .NET, Python) can communicate using SOAP.

ii) WSDL (Web Services Description Language)

Concept in Brief

WSDL is an **XML document** that describes what a web service does and how to use it.

Explanation

1. Defines Web Service Details

- What functions the service provides
- What input and output messages look like
- Where the service is located (URL)
- How to call it

2. Written in XML

- Platform and language independent.

3. Main Parts of WSDL

- **Types** → Data types used
- **Message** → Input/output data
- **Port Type** → Operations or functions
- **Binding** → Protocol used (SOAP)
- **Service** → Endpoint URL

4. Used for Automatic Client Generation

- Tools like Apache Axis, JAX-WS can create client-side code automatically from WSDL.

5. Essential for SOAP Web Services

- Without WSDL, client does not know how to call the service.

Q4 (a) Explain Struts framework with its components. Also explain interceptors. (9 Marks)

Ans:

Struts is a **Java-based MVC framework** used to build web applications. It separates application logic into **Model**, **View**, and **Controller**.

Struts Framework – Explanation

1. Struts is MVC Framework

- Struts follows **Model-View-Controller** architecture.
- It helps to separate:
 - **View (JSP UI)**
 - **Controller (ActionServlet)**
 - **Model (Business logic, Java classes)**

Main Components of Struts Framework

1. ActionServlet (Controller)

- It is the **main controller** of the application.
- Receives all requests from users.
- Reads configuration from struts-config.xml.
- Calls the appropriate Action class.

2. ActionForm (Form Bean)

- Java class used to **capture user input** from JSP.
- Validates the input.
- Example: LoginForm, RegisterForm.

3. Action Class

- Contains **business logic**.
- For each request, a separate Action class is created.
- Returns the result as **ActionForward**.

4. ActionForward

- Decides **which page to display** next.
- Example: success.jsp, error.jsp.

5. struts-config.xml

- The **heart** of the framework.
- Contains mappings of:
 - ActionServlet
 - Form Beans
 - Action classes
 - Forward pages

6. JSP Pages (View)

- Used for displaying UI.
- They show results received from Action classes.

🔗 Flow of Struts Application

1. User sends request → goes to **ActionServlet**.
2. ActionServlet picks matching **ActionForm**.
3. User data stored in ActionForm.
4. ActionServlet calls appropriate **Action class**.
5. Action class executes logic and returns **ActionForward**.
6. JSP page displays the output.

Interceptors in Struts

Concept

An **Interceptor** is a component that performs some operation *before* or *after* an Action class is executed.

Examples of Interceptor Tasks

- Checking if user is logged in
- Logging request data
- Validation
- File upload processing
- Exception handling

How Interceptors Work (Point-to-Point)

1. Request received by framework
2. Interceptor executes **before** Action
3. Action executes
4. Interceptor executes **after** Action
5. Response sent to browser

Advantages

- Reusable
- Clean and modular code
- No need to write repetitive code in every Action class

Q4 (b) Explain JSP support for MVC (Model–View–Controller) for developing web applications.(8)

Ans:

MVC is a design pattern that separates **user interface**, **data**, and **logic**. JSP is mainly used as **View**, but it supports all MVC components.

MVC Components Using JSP

1. Model (Data + Business Logic)

- Model = JavaBeans, POJO classes, Database code.
- Handles:
 - Data processing
 - Database transactions
 - Application logic
- JSP interacts with Model using:
 - JSTL
 - EL (Expression Language)
 - <jsp:useBean> tag

2. View (User Interface) — JSP

- JSP acts as **View**.
- Displays output from the Model.
- Uses:
 - HTML
 - CSS
 - JavaBeans
 - JSP EL and JSTL
- Does **not** contain business logic.

3. Controller (Servlet)

- Servlet acts as the **Controller**.
- Controller tasks:
 - Accepts user request
 - Calls the correct Model components
 - Stores results in request/session
 - Forwards to JSP (View)

Uses RequestDispatcher:

```
request.getRequestDispatcher("result.jsp").forward(request, response);
```

How JSP Fits into MVC Flow

Step-by-Step MVC Flow

1. **User sends request** (e.g., login).
2. Request goes to **Servlet (Controller)**.
3. Servlet calls **Model** to process the request.
4. Model returns data/results.
5. Servlet forwards result to the **JSP page (View)**.
6. JSP shows the data to the user.

Benefits of Using JSP in MVC

- Clear separation of roles
- Easy maintenance
- JSP contains only UI → cleaner code
- Servlet handles logic → improved performance
- Model stores data → reusable classes

Q5 a) Identify and explain steps involved in connecting to MySQL with PHP.[9]

Ans:

PHP connects to MySQL using functions like **mysqli_connect()** or **PDO**, allowing web pages to store, update and retrieve data dynamically.

Steps 1) Install & Start MySQL Server

- MySQL must be installed and running in the background.
- Create a database (example: student_db).

2) Configure PHP with MySQL Extension

- PHP must have mysqli or PDO enabled.
- Usually enabled by default in XAMPP/WAMP.

3) Establish Connection

- Use mysqli_connect(host, username, password, database).

```
$conn = mysqli_connect("localhost", "root", "", "student_db");
```

- If connection fails, an error is shown.

4) Check Connection Status

```
if (!$conn) {  
    die("Connection failed: " . mysqli_connect_error());  
}
```

- This ensures the program stops if there is an error.

5) Execute SQL Queries

- Queries like INSERT, SELECT, UPDATE, DELETE.

```
$result = mysqli_query($conn, "SELECT * FROM students");
```

6) Process the Result

- Fetch rows from result.

```
while($row = mysqli_fetch_assoc($result)){  
    echo $row['name'];  
}
```

7) Close the Connection

```
mysqli_close($conn);
```

- This frees server memory.

Q5 b) Short Notes (9 Marks)

Ans:

i) Overview of ASP.NET

- ASP.NET is a **server-side web development framework** by Microsoft.
- It allows developers to build dynamic websites, APIs and web applications.
- Runs on the **.NET Framework / .NET Core**.
- Supports multiple languages: **C#, VB.NET**.
- Uses rich controls like **TextBox, Button, GridView, DropDownList**.
- Provides strong features such as:
 - Built-in security
 - Automatic memory management
 - Session & state management
 - MVC architecture
 - Web services & API development
- Output is converted into HTML and sent to the client browser.

ii) Overview of C#

- C# (C-Sharp) is a **modern, object-oriented programming language** by Microsoft.
- Used widely in ASP.NET, desktop apps, game development (Unity), mobile apps (Xamarin).
- Combines features of Java and C++.
- Supports OOP concepts:
 - Classes
 - Objects
 - Inheritance
 - Polymorphism
 - Encapsulation
- Easy syntax, type-safe, automatic garbage collection.
- Works seamlessly with .NET Framework and Visual Studio IDE.

OR

Q6 a) Explain in detail WAP Architecture & WML.[9]

Ans:

WAP (Wireless Application Protocol) is a standard for delivering internet content to mobile devices. WML (Wireless Markup Language) is the language used to create WAP pages.

1) WAP Architecture

(a) Mobile Device

- User interacts through mobile phone (WAP-enabled).

(b) WAP Micro Browser

- Special browser capable of reading **WML pages**.

(c) WAP Gateway / Proxy

- Converts **WML ↔ HTML**
- Converts **WAP requests ↔ HTTP requests**
- Compresses data to reduce size.

(d) Internet / Web Server

- Hosts actual web applications.
- Responds to WAP gateway via HTTP/HTTPS.

(e) Application Server

- Executes business logic (e.g., PHP, ASP.NET, Java).

Flow of WAP Request

1. User sends request from mobile browser.
2. WAP Gateway converts request to HTTP.
3. HTTP request goes to web server.
4. Server sends back data to gateway.
5. Gateway converts it to WML.
6. WML sent back to mobile phone.

2) WML (Wireless Markup Language)

- Similar to HTML but designed specially for **small screens and low bandwidth devices**.
- Uses **deck** and **cards** structure (like pages inside book).
- Supports navigation, text, links and simple UI.

Sample WML Code

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.1//EN">
<wml>
  <card id="welcome" title="Home">
    <p>Welcome to WML Page</p>
  </card>
</wml>
```

Q6 b) Explain functions in PHP with example & session management.[9]

Ans:

Functions are reusable blocks of code. Session management stores user data across multiple web pages.

Functions in PHP (with example)

Definition

A function is a block of statements that performs a specific task and can be reused.

Syntax

```
function functionName(parameters) {  
    // statements  
}
```

Example

```
function add($a, $b){  
    return $a + $b;  
}
```

```
echo add(10, 20); // Output: 30
```

Benefits

- Reusability
- Easy debugging
- Code modularity
- Better readability

Session Management in PHP

What is a Session?

- A session stores information for a user across multiple pages.
- Example: login credentials, shopping cart, user settings.

Starting a Session

```
session_start();
```

Storing Data in Session

```
$_SESSION['username'] = "Vijay";
```

Accessing Session Data

```
echo $_SESSION['username'];
```

Destroying Session

```
session_destroy();
```

Q7 (a) Explain classes and objects in Ruby with appropriate examples. [9]

Ans:

In Ruby, **everything is an object**, and objects are created from **classes**. A **class** is a blueprint, while an **object** is a real instance created from that blueprint.

1. Class in Ruby

A class groups together:

- Data (variables)
- Behaviour (methods)

Syntax of a class

```
class ClassName
  # variables and methods
end
```

2. Creating an object

Objects are created using the .new method.

```
obj = ClassName.new
```

Example: Simple Ruby Class & Object

```
class Student
  def initialize(name, marks)
    @name = name
    @marks = marks
  end

  def display
    puts "Name: #{@name}"
    puts "Marks: #{@marks}"
  end
end
```

```
# Creating objects
s1 = Student.new("Ravi", 85)
s2 = Student.new("Anita", 90)
```

```
s1.display
s2.display
```

Explanation

- initialize → constructor, runs automatically when object created
- @name and @marks → instance variables
- display → method
- Student.new → creates objects

This is how Ruby uses classes and objects to organize code in an OOP way.

Q7 (b) Introduce the concept of Rails application. Describe layouts & stylesheet in Rails. [8]

Ans:

1. Concept of Rails Application

Ruby on Rails is a **web application framework** based on:

- MVC (Model–View–Controller)
- Convention over Configuration
- DRY (Don't Repeat Yourself)

A Rails application contains:

- **Models** → handle database logic
- **Views** → HTML pages seen by user
- **Controllers** → connect models & views
- **Routes** → decide URL paths
- **Assets** → CSS, JS, images

Rails helps to quickly build **dynamic, database-driven websites**.

2. Layouts in Rails

A **layout** is like a master template used for common design across pages.

Example: header, footer, navbar, sidebar.

Default layout file:

app/views/layouts/application.html.erb

This file contains:

```
<!DOCTYPE html>
<html>
<head>
  <title>MyApp</title>
  <%= csrf_meta_tags %>
  <%= csp_meta_tag %>
  <%= stylesheet_link_tag "application" %>
</head>
<body>

<%= yield %>  <!-- Page content goes here -->

</body>
</html>
```

- yield → inserts the page's actual content
- Prevents repeating same HTML on every page

3. Stylesheets in Rails

Rails uses the **asset pipeline** to manage CSS.

Stylesheets are stored in:

app/assets/stylesheets/

Example file: application.css

You include it in layout using:

```
<%= stylesheet_link_tag "application" %>
```

You can add your own CSS files:

app/assets/stylesheets/home.css

Rails compiles and optimizes all CSS before sending to the browser.

OR-Q8 (a) Explain the scalar types & their operations in Ruby. [8]

Ans:

Scalar types in Ruby are **basic single-value types**.

1. Integer

Whole numbers

Operations: +, -, *, /, %, **

Example:

```
a = 10
```

```
b = 3
```

```
puts a + b
```

```
puts a % b
```

2. Float

Decimal numbers

Example:

```
x = 5.5
```

```
y = 2.0
```

```
puts x / y
```

3. String

Sequence of characters

Operations:

- Concatenation: "Hello" + " World"
- Repetition: "Hi" * 3
- Length: "Ruby".length

Example:

```
name = "Ruby"
```

```
puts name.upcase
```

4. Boolean

true or false

Used in conditions.

5. Nil

Represents "nothing" or "no value".

6. Pattern Matching (Regular Expressions)

Used using /pattern/.

Example:

```
puts "hello" =~ /he/ # returns index 0
```

Q8 (b) Explain Architecture of EJB & explain types of EJB in detail. [9]

Ans:

What is EJB?

EJB (Enterprise JavaBean) is a **server-side component** used to develop:

- Large-scale
- Secure
- Transaction-based applications in Java.

Architecture of EJB

EJB architecture has **three main parts**:

1. Client Layer

User interface or application that calls EJB.

2. EJB Container

Runs on application server.

Provides:

- Security
- Transactions
- Life-cycle management
- Object pooling

3. EJB Server

The actual application server (JBoss, GlassFish, WebLogic).

Types of EJB

1. Session Beans

Perform business logic.

a) Stateless Session Bean

- Does not store client data
- Example: login check, calculations

b) Stateful Session Bean

- Stores data of one client session
- Example: shopping cart

c) Singleton Session Bean

- Only one instance in entire application
- Example: configuration, logs

2. Message-Driven Beans (MDB)

Used for asynchronous communication.

Works with JMS (Java Messaging Service).

Example: sending notifications, processing messages in queue.

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